Title

A Computational Workflow for Façade-Level Solar Energy Assessment Using Solar Geometry, Urban Shading, and Cloud-Adjusted Irradiance

Abstract

Photovoltaic systems are traditionally deployed on horizontal or near-horizontal building surfaces, limiting their applicability in dense urban environments where roof area is constrained. This study presents a computational workflow for assessing solar energy generation on vertical building façades. The workflow evaluates façade-level solar output across four cardinal orientations (North, South, East, West) for buildings located in New York City, Pune, and Paris, representing diverse latitudes and climatic conditions. By integrating solar position modeling, geometry-based urban shading analysis, and cloud-adjusted irradiance data from the SoDa/NSRDB framework, the method estimates time-resolved façade energy production at 30-minute intervals.

1. Introduction

Façade performance is strongly influenced by orientation, surrounding urban geometry, and atmospheric conditions. Thus, the workflow considers contextual information, follows a sequence of steps shown in Fig-1.

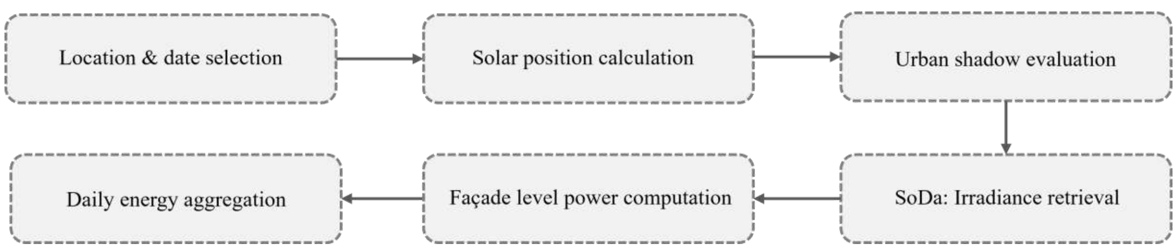

Fig-1 Workflow Sequence

2. Methodology

2.1 Study Locations and Scope

Users select between 3 locations 1) NYC, USA 2) Pune, India 3) Paris, France, through toggles designed as per Fig-2. For each location, a representative building is analyzed with façades aligned to the four cardinal directions. Parameters that affect the solar energy generation are Location, time, date, number of buildings, building height, width, length position.

2.2 Solar Geometry and Sun Position

The Sun path simulation seen in Fig-3 is computed using deterministic Earth-Sun geometry, deriving solar azimuth and altitude from latitude, date, and time via standard spherical trigonometric relationships, following established formulations in solar engineering literature (Duffie and Beckman, 2013).

Fig-2 Toggle Interface

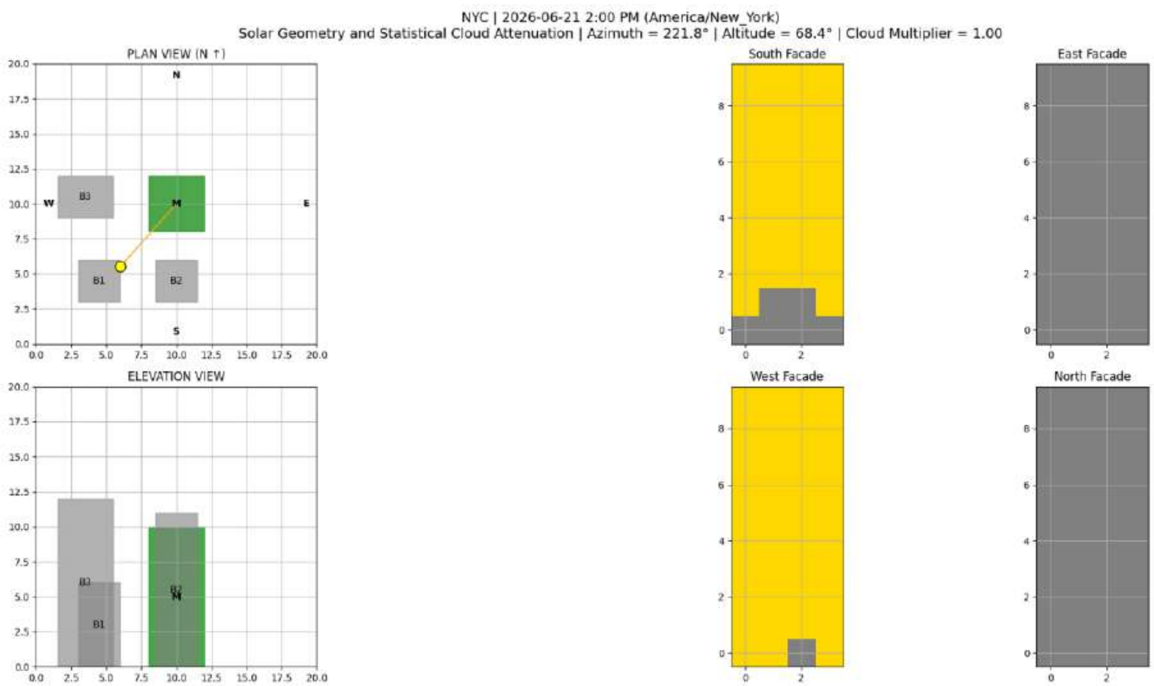

Fig- 3 Sun Path & building facade simulation

2.3 Urban Shading and Cloud-Aware Façade Irradiance Modeling

To account for both geometric and atmospheric effects on façade-level solar availability, a combined urban shading and cloud-aware irradiance model is implemented. Surrounding buildings are represented as simplified rectangular volumes defined by plan dimensions and height, enabling a geometry-based evaluation of inter-building shading. Each façade is discretized into a regular grid of surface elements, and at every simulation time step, ray-based geometric checks using the computed solar azimuth and altitude determine whether individual elements are sunlit or shaded, producing a binary shadow mask per façade.

Atmospheric variability is incorporated using cloud-adjusted irradiance data accessed through the SoDa framework, which leverages datasets derived from the National Solar Radiation Database (NSRDB). SoDa provides irradiance values at a 30-minute temporal resolution for historical reference years (2016–2017) and enables stochastic mapping of cloud-induced variability beyond this period. In this workflow, a cloud attenuation multiplier computed by comparing measured irradiance to corresponding clear-sky estimates is applied uniformly across sunlit façade elements at each time step, yielding an effective façade-level irradiance field that reflects both urban occlusion and statistically representative atmospheric conditions (Losada et al., 2020).

2.4 Energy Computation

For each façade, the active (sunlit) surface area is obtained from the binary shadow mask generated by the urban shading model and multiplied by the cloud-adjusted global irradiance retrieved via the SoDa/NSRDB framework. Instantaneous electrical power is calculated as:

$$P(t) = A(t) \times I(t) \times \eta \times \cos(\theta_v)$$

where $A(t)$ is the sunlit façade area, $I(t)$ is the cloud-adjusted irradiance, η is the assumed photovoltaic conversion efficiency (15%, representative of commercially available silicon PV modules), and $\cos(\theta_v)$ accounts for the vertical projection of solar incidence based on solar altitude. Instantaneous power is computed independently for each façade orientation and aggregated to obtain total building power. Daily energy generation is obtained by temporally integrating the power time series over 30-minute intervals, yielding façade-wise and total daily energy in kWh. This formulation follows standard plane-of-array solar energy modeling practices and is consistent with established treatments of photovoltaic power estimation in solar engineering literature (Duffie & Beckman, 2013).

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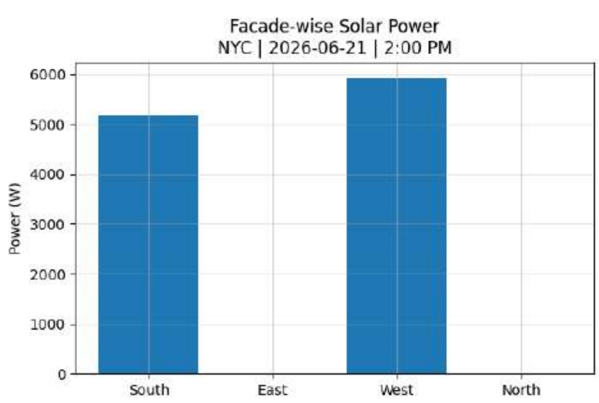

Fig-4 Instantaneous power

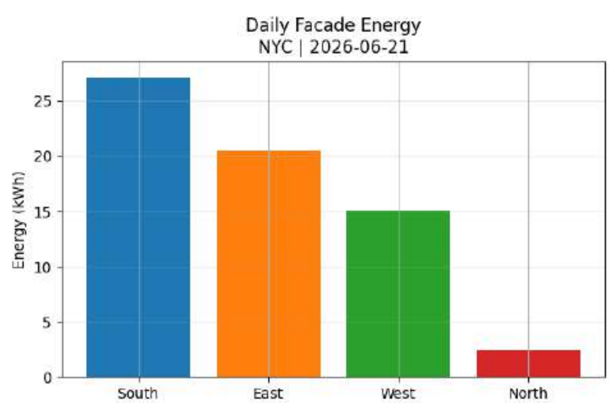

Fig-5 Comparison of various facades for a single day

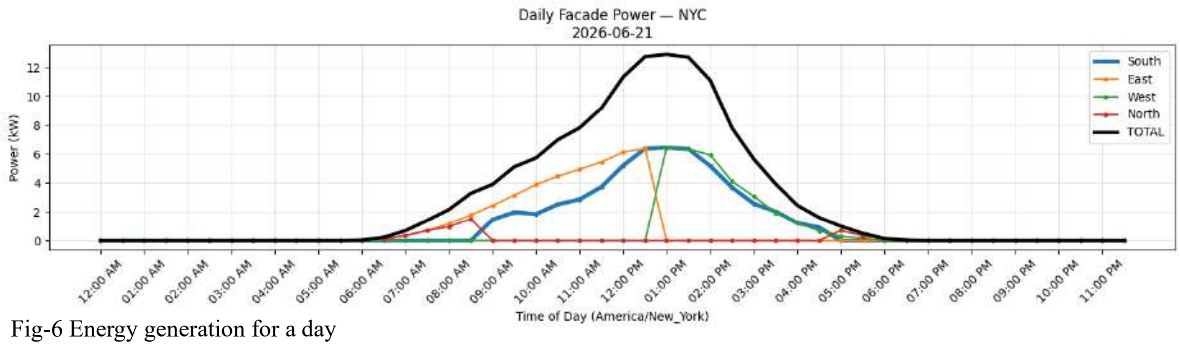

Fig-6 Energy generation for a day

The façade-level power and energy outputs are visualized using three representations. Instantaneous power plot, Fig- 4 shows orientation-wise electrical output at a selected time, reflecting the combined effects of solar geometry and shading. Daily power curves, Fig-6, illustrate time-resolved fading over a full day, capturing orientation-dependent exposure and urban occlusion effects. A daily energy bar chart, Fig-5 summarizes total energy generation per façade by integrating power over time, enabling direct comparison of relative façade performance

3. Limitations and Assumptions

The workflow relies on irradiance data from NSRDB reference years (2016–2017), with future dates represented through SoDa’s stochastic mapping to historical patterns. Building geometry is simplified, assuming small-scale rectangular plans (4 m × 5 m), planar façades aligned to cardinal directions, and uniform photovoltaic coverage without accounting for openings or articulation, which results in modest energy outputs as shown in Fig. 6. Accordingly, the model is intended for comparative analysis and early-stage feasibility assessment rather than detailed performance prediction, with future work enabling larger building scales and location-specific data for more realistic outcomes.

4. Conclusion

The example case shows that vertical façade energy generation depends more on orientation and solar exposure than on surface area alone. South-, east-, and west-facing façades generate significantly more energy than the north façade, with east and west façades benefiting from extended morning and afternoon exposure. This demonstrates the strong orientation sensitivity of vertical photovoltaics and illustrates how the proposed workflow supports façade-level energy assessment in dense urban contexts.

Reference

1. Duffie, J. A., & Beckman, W. A. (2013). *Solar engineering of thermal processes* (4th ed.). Hoboken, NJ: John Wiley & Sons.
2. Losada, I., Kazemtabrizi, B., Venkataramanan, G., & Scaglione, A. (2020). *SoDa: An irradiance-based synthetic solar data generation tool*. IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm).